

Internship Report

An Agent Based Market Model
on Sustainable Aviation Fuel

AE5051: Internship Master Aerospace Engineering
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Delft University of Technology

Internship Report

An Agent Based Market Model on Sustainable Aviation Fuel

by

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Preface

A preface...

*Jelle Weijland
Delft, December 2025*

Summary

context

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Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m ³]
...		

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Introduction

1.1. Context

Sustainable Aviation Fuel (SAF) is a key technology for reducing aviation emissions, offering a drop-in replacement for conventional jet fuel with lifecycle CO₂ reductions of up to 80%. Because long-haul aviation depends on high energy-density fuels, SAF is expected to play a central role in decarbonizing the sector over the coming decades.

SAF can be produced from several feedstocks such as used cooking oil, agricultural residues, forestry waste, and municipal solid waste. Current production is dominated by HEFA, while alternative pathways such as Fischer–Tropsch, Alcohol-to-Jet, and Power-to-Liquid are progressing toward commercial scale.

Global mandates are increasing rapidly. The EU requires 6% SAF blending by 2030 and 70% by 2050, while the United States targets 3 billion gallons of annual SAF production by 2030. India has announced 1% blending by 2027 and 5% by 2030, and China is developing similar policies. These targets imply a steep increase in future demand.

Feedstock availability is a major constraint. India alone produces roughly 500 million tons of agricultural residue each year, with about 140 million tons available for energy use. SAF production typically requires 1.5–3 tons of feedstock per ton of SAF. For the EU's 2030 mandate, this translates to an estimated 7.6 million tons of feedstock annually, rising to about 88 million tons by 2050.

These requirements create a strategic dynamic between SAF investors and feedstock aggregators, who control supply chains and often have profitable alternative uses for biomass. Understanding this interaction is essential for analyzing long-term SAF market development.

1.2. Model Overview

1.2.1. Agent based Modeling

1.2.2. Agents

Feedstock Aggregator

SAF Production Plant

Investor

The baseline SAF Market Model simulates an emerging SAF industry with three agent types: feedstock aggregators, SAF production sites, and investors. The model runs in yearly timesteps.

Feedstock aggregators represent regional farmers with biomass supply. Each aggregator has a feedstock price and a maximum yearly availability. At the start of each year, the available feedstock is sampled using a random multiplier, resulting in variation between good and poor supply years.

SAF production sites convert biomass into fuel. Sites have fixed capacity and purchase all feedstock directly from their state's price. Their Short-Run Marginal Cost (SRMC) consists of feedstock cost,

operational expenditure, transport cost and a small profit margin. The plants SRMC is put into an merit order in which the last plant filling the demand marginally is used as market price setter.

Investors decide to build new sites or not. They evaluate each investment using an NPV (Net Present Value) calculation. Each investor has an individual initial discount rate (sampled between 6–12%) and an optimism factor that shapes expected future prices. Investors select the state with the lowest feedstock cost and invest if the calculated Net Present Value, which is calculated by expected ebit over some discount rate dependent factor, of the specific plant that is being evaluated, is positive.

Each simulation year follows a fixed structure. Aggregators update biomass availability, production sites operate within their physical and supply limits and market prices are determined through merit-order pricing. The model also generates a long-term price forecast by simulating future demand growth and capacity additions, which is shared with investors and is used in the NPV calculation.

After each year, investors evaluate the performance of their existing sites using a so called ROACE (Return on Average Capital Employed). Poor returns increase their discount rate, while strong performance reduces it, modelling changes in risk perception. New potential investors may also enter the market each year, expanding the pool of available capital.

Because all feedstock is purchased on the fixed price market, the baseline model exposes production sites fully to price and supply volatility. States with low feedstock prices offer a structural cost advantage, while states with unstable supply experience higher risk. The interaction of biomass variability, merit-order pricing, and investment behaviour creates dynamic feedback loops that shape market development without the need for central planning.

1.3. Research Question

This research examines how feedstock should be priced to support efficient SAF market development while ensuring fair returns for both investors and aggregators. Simple supply–demand pricing does not fully capture the realities of biomass markets, where aggregators face rising marginal costs, infrastructure limits, and opportunity costs from competing feedstock uses. A more structured pricing approach is therefore required.

The objectives of this work are:

1. Develop and evaluate a pricing mechanism that more accurately reflects feedstock cost structures than traditional market-clearing prices.
2. Compare market outcomes under tier-based and simple pricing, including investment timing, capacity expansion, price evolution, and profitability.
3. Assess the sensitivity of market dynamics to key parameters such as costs, capacities and ROACE.
4. Provide behavioural insights to support effective SAF market development.

1.4. Challenges and Motivation

Traditional supply–demand pricing assumes that feedstock aggregators are passive price-takers willing to sell biomass to any buyer. In reality, aggregators operate across multiple markets and often have profitable alternatives such as selling to agriculture, supplying bio-energy producers, or processing the biomass themselves. This flexibility gives them substantial bargaining power and means that additional SAF demand does not translate directly into lower prices or unlimited supply.

Serving more SAF plants also increases aggregator costs. Early plants benefit from favourable locations with short transport distances and established logistics, while later plants require longer hauls, additional storage, and more complex infrastructure. Existing collection systems can efficiently supply initial volumes, but scaling up requires capital investments in expanded collection networks and processing capacity.

Aggregators also face opportunity costs when committing large volumes to SAF. Long-term contracts limit their ability to respond to price signals from other markets, especially during periods of high agricultural commodity prices. In addition, supplying multiple plants exposes aggregators to higher risk

from weather variability, harvest uncertainty, and regulatory changes, which justifies a risk premium in pricing.

These factors support a tier-based pricing structure in which costs rise as more capacity is served. This approach more accurately reflects the economics of biomass supply and creates realistic competitive dynamics among investors.

An introduction... [1]

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Methodology

2.1. Tier Mechanism

The tier mechanism organizes feedstock supply into discrete capacity blocks, each with a fixed annual volume and an associated unit cost. This structure captures the rising marginal costs faced by aggregators as they serve additional SAF plants, reflecting infrastructure limitations, longer transport distances, and expanding collection networks.

In the baseline implementation, each tier provides 120,000 tons of feedstock per year, while each SAF plant requires 100,000 tons to operate at full capacity. This deliberate mismatch creates realistic competitive dynamics: multiple plants contracting with the same aggregator will push the system into higher-cost tiers.

Tier Structure:

- Tier 1: 120,000 tons at \$500/ton (existing infrastructure)
- Tier 2: 120,000 tons at \$700/ton (expanded collection network)
- Tier 3: 120,000 tons at \$900/ton (distant sources, higher logistics cost)

Numerical Example

Plant 1 enters:

- Contracts for 100,000 tons from Tier 1 at \$500/ton.
- Aggregator cost: \$50 million.
- Aggregator revenue: \$50 million.
- Profit: \$0 (cost recovery).
- Remaining Tier 1 capacity: 20,000 tons.

Plant 2 enters:

- Needs 100,000 tons; receives 20,000 tons from Tier 1 and 80,000 tons from Tier 2.
- Aggregator total cost:
 - Tier 1: 120,000 tons × \$500 = \$60M
 - Tier 2: 80,000 tons × \$700 = \$56M
- Total cost: \$116M for 200,000 tons.
- Zero-profit equilibrium price: $\$116\text{M} \div 200,000 = \text{\$580/ton}$.
- Both Plant 1 and Plant 2 now pay \$580/ton.

Plant 3 enters:

- Receives 40,000 tons from Tier 2 and 60,000 tons from Tier 3.
- Aggregator total cost:
 - Tier 1: $120,000 \times \$500 = \60M
 - Tier 2: $120,000 \times \$700 = \84M
 - Tier 3: $60,000 \times \$900 = \54M
- Total cost: \$198M for 300,000 tons.
- New equilibrium price: $\$198\text{M} \div 300,000 = \$660/\text{ton}$.

Resulting Dynamics

Erosion of first-mover advantage: Early entrants initially benefit from lower prices, but each new plant triggers a recalculation that raises the equilibrium price for all buyers.

Cross-subsidization: At equilibrium, early plants pay above their marginal cost and subsidize later entrants; late entrants pay below their marginal cost. This ensures aggregators break even on average.

Locked-in cost disadvantage: Plants entering after higher tiers are activated permanently face higher feedstock costs, influencing long-term competitiveness.

Strategic timing incentives: Investors must balance entering early (lower feedstock costs) against waiting for more market information—creating meaningful timing trade-offs.

2.1.1. Contract Structure

2.1.2. Take-or-Pay Mechanism

2.1.3. Investor Decision Making

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Results

- 3.1. Market Price Development
- 3.2. Supply and Demand
- 3.3. Price Forecast
- 3.4. Owned sites per investor
- 3.5. Production per Plant vs Demand
- 3.6. Curtailed Volume & Penalty Cost
- 3.7. Investor KPI's
- 3.8. SAF Supply by Plant
- 3.9. Feedstock Availability per State
- ...

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Sensitivity Analysis

- 4.1. Tier cost Increment variation - Tier Volume increase effect on investment incentive by means of measuring number of second or more investments
- 4.2. Tier cost Increment variation - Tier Volume increase effect on curtailed volume
- 4.3. Tier cost Increment variation - Tier Volume increase effect on
....

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Discussion

5.1. Key Findings

5.2. Implications

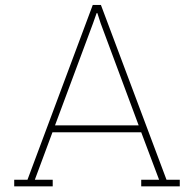
5.3. Limitations

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Conclusion and Recommendations

References

- [1] I. Surname, I. Surname, and I. Surname. “The Title of the Article”. In: *The Title of the Journal* 1.2 (2000), pp. 123–456.



Source Code Example

Adding source code to your report/thesis is supported with the package listings. An example can be found below. Files can be added using \lstinputlisting[language=<language>]{<filename>}.

```
1 """
2 ISA_Calculator: import the function, specify the height and it will return a
3 list in the following format: [Temperature, Density, Pressure, Speed of Sound].
4 Note that there is no check to see if the maximum altitude is reached.
5 """
6
7 import math
8 g0 = 9.80665
9 R = 287.0
10 layer1 = [0, 288.15, 101325.0]
11 alt = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000]
12 a = [-.0065, 0, .0010, .0028, 0, -.0028, -.0020]
13
14 def atmosphere(h):
15     for i in range(0, len(alt)-1):
16         if h >= alt[i]:
17             layer0 = layer1[:]
18             layer1[0] = min(h, alt[i+1])
19             if a[i] != 0:
20                 layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
21                 layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
22             else:
23                 layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
24     return [layer1[1], layer1[2]/(R*layer1[1]), layer1[2], math.sqrt(1.4*R*layer1[1])]
```